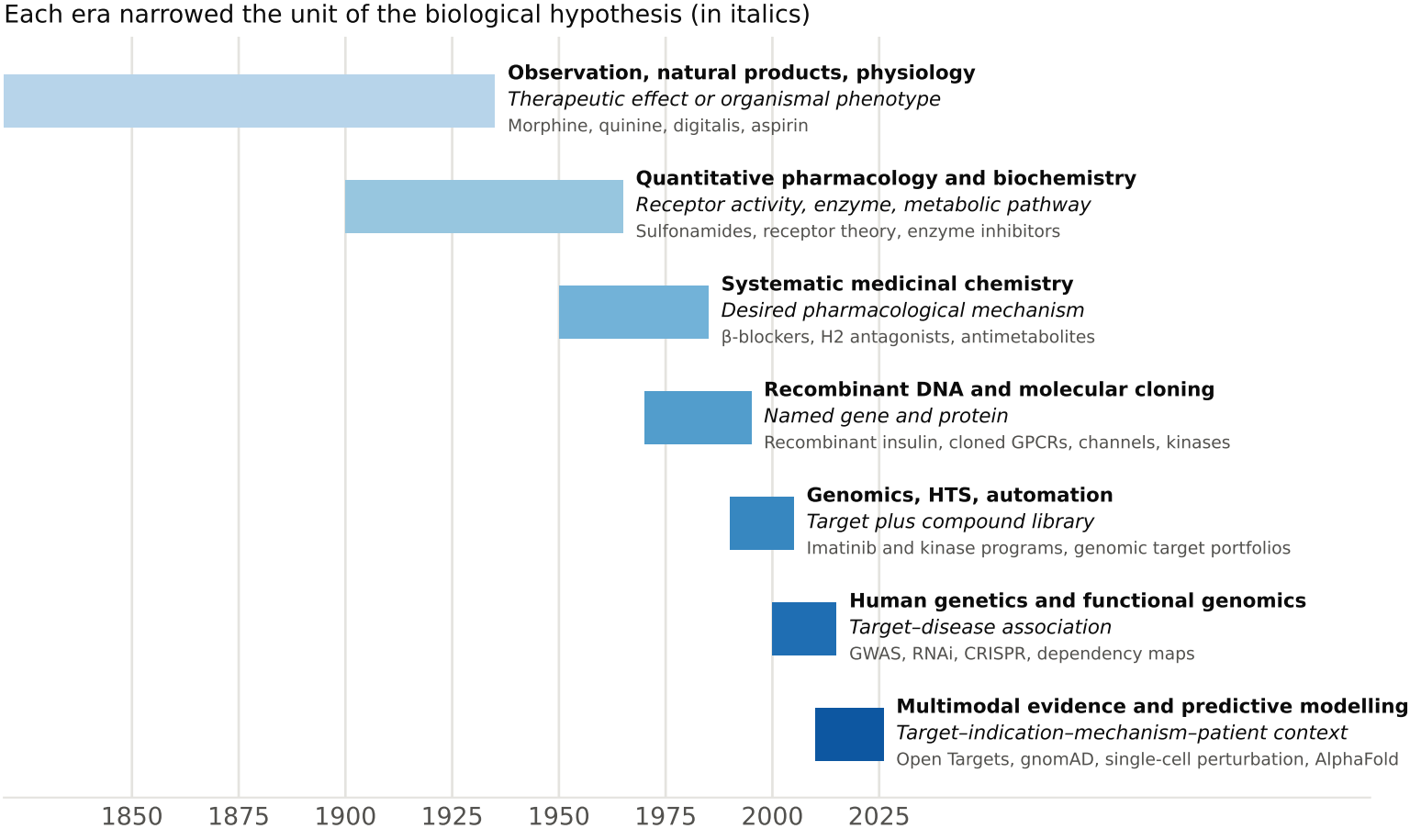
Each era narrowed the unit of the biological hypothesis (in italics)
Observation, natural products, physiology
Therapeutic effect or organismal phenotype
Morphine, quinine, digitalis, aspirin
Quantitative pharmacology and biochemistry
Receptor activity, enzyme, metabolic pathway
Sulfonamides, receptor theory, enzyme inhibitors
Systematic medicinal chemistry
Desired pharmacological mechanism
β-blockers, H2 antagonists, antimetabolites
Recombinant DNA and molecular cloning
Named gene and protein
Recombinant insulin, cloned GPCRs, channels, kinases
Genomics, HTS, automation
Target plus compound library
Imatinib and kinase programs, genomic target portfolios
Human genetics and functional genomics
Target–disease association
GWAS, RNAi, CRISPR, dependency maps
Multimodal evidence and predictive modelling
Target–indication–mechanism–patient context
Open Targets, gnomAD, single-cell perturbation, AlphaFold
1850
1875
1900
1925
1950
1975
2000
2025